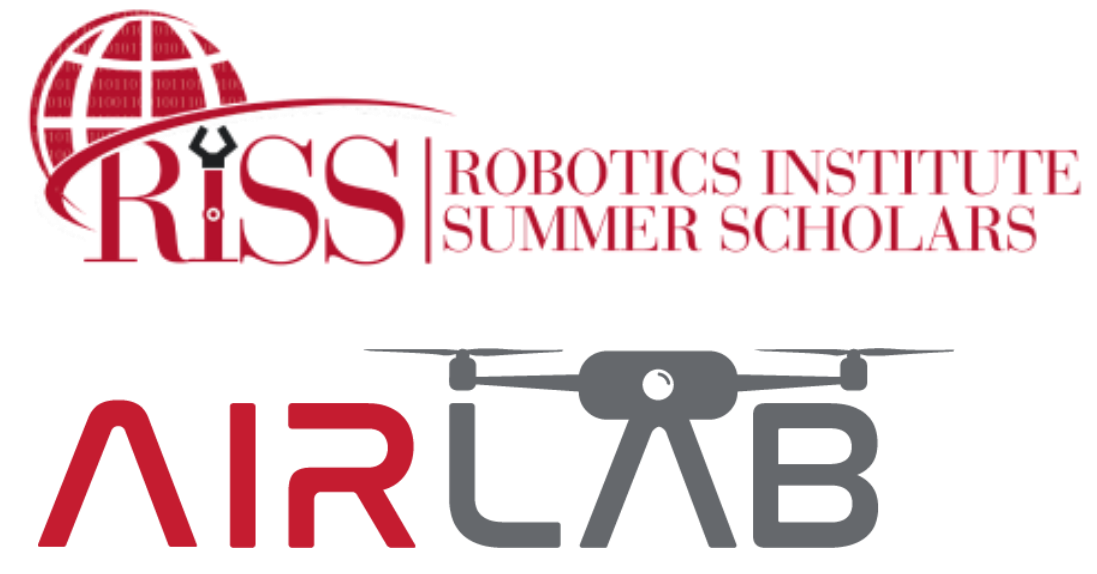


3D Gaussian Splatting for Learned Drone Navigation



Daniel Kim¹, Yikuan Fang², Wenshan Wang², Sebastian Scherer²

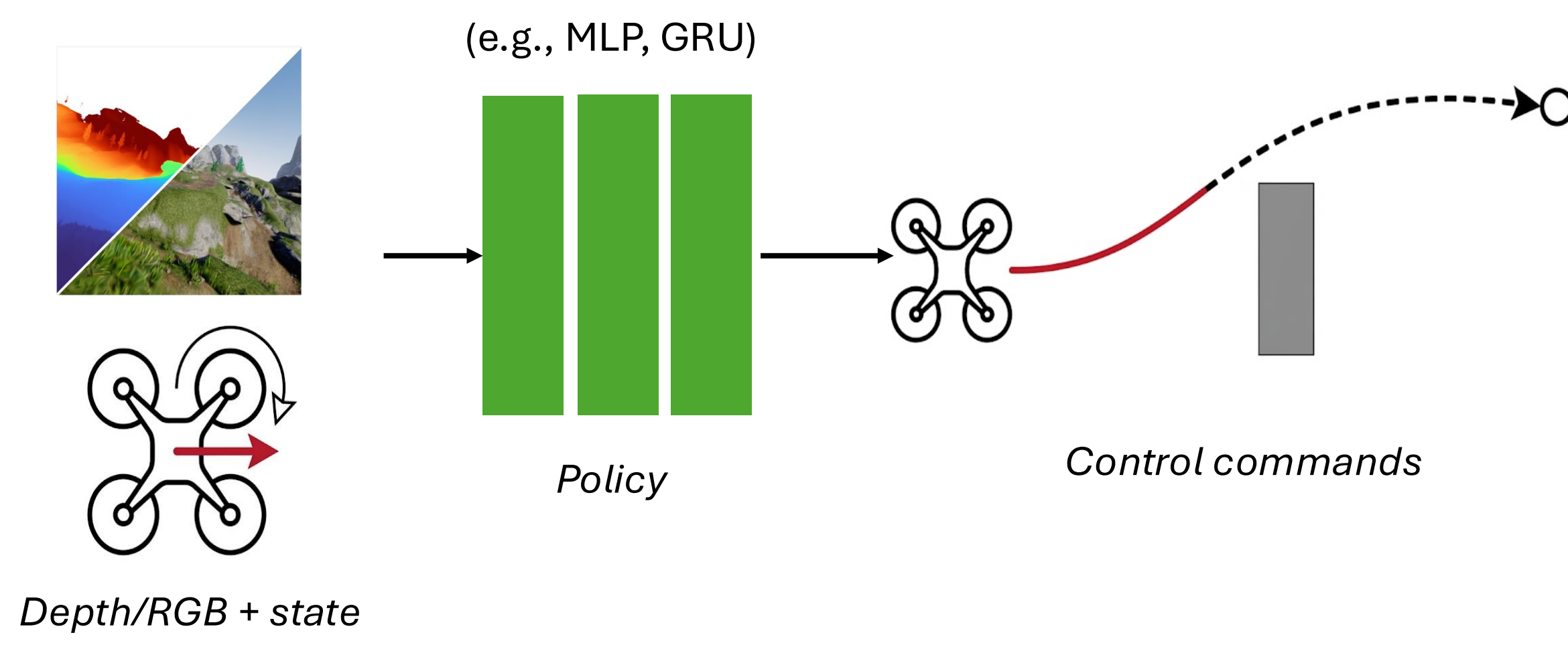
¹Rice University

²Carnegie Mellon University



Introduction

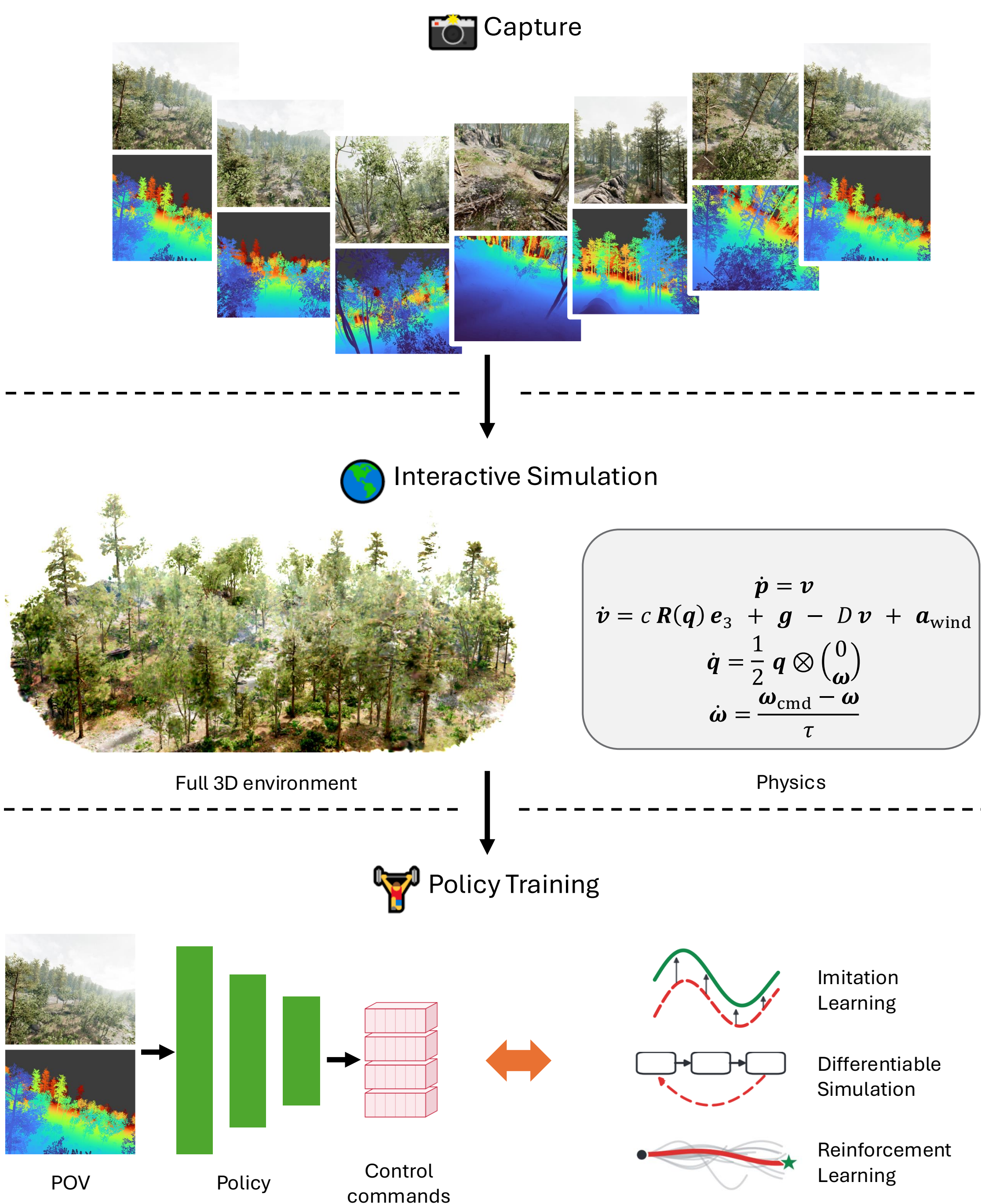
Goal: lightweight end-to-end navigation policy for agile drone flight in the wild.



Current policies use **depth** because it is appearance-invariant. But depth sensors add payload, power, and cost.

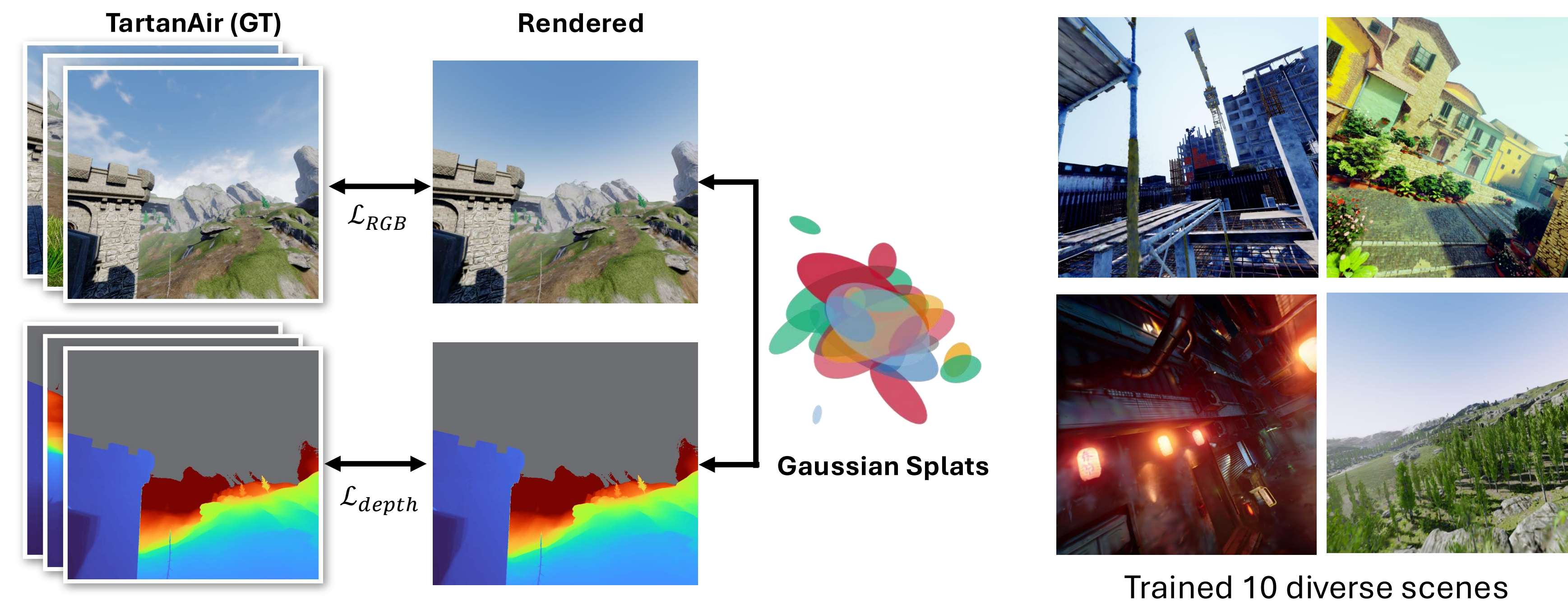
Contribution: A simulator to train a monocular RGB policy on a diverse set of scenes.

Simulator

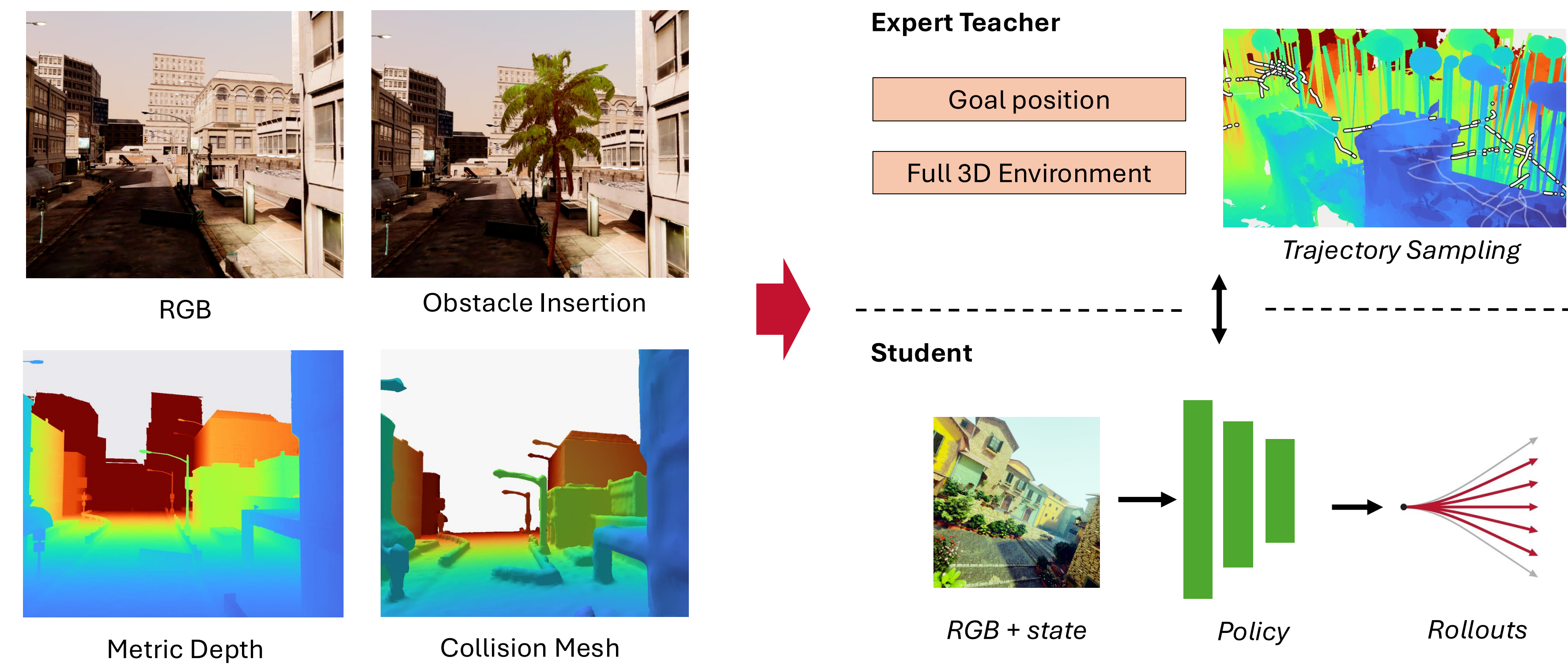


Methods

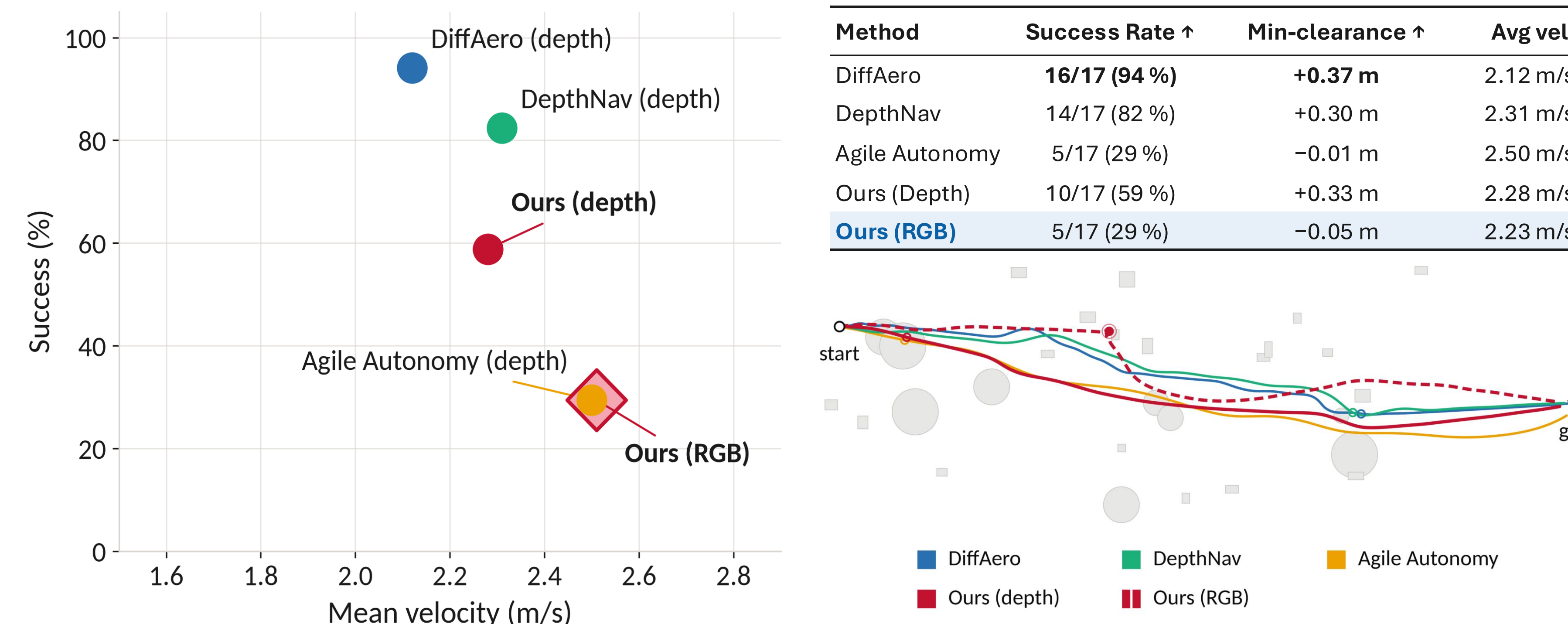
Use 3DGS as a simulation environment



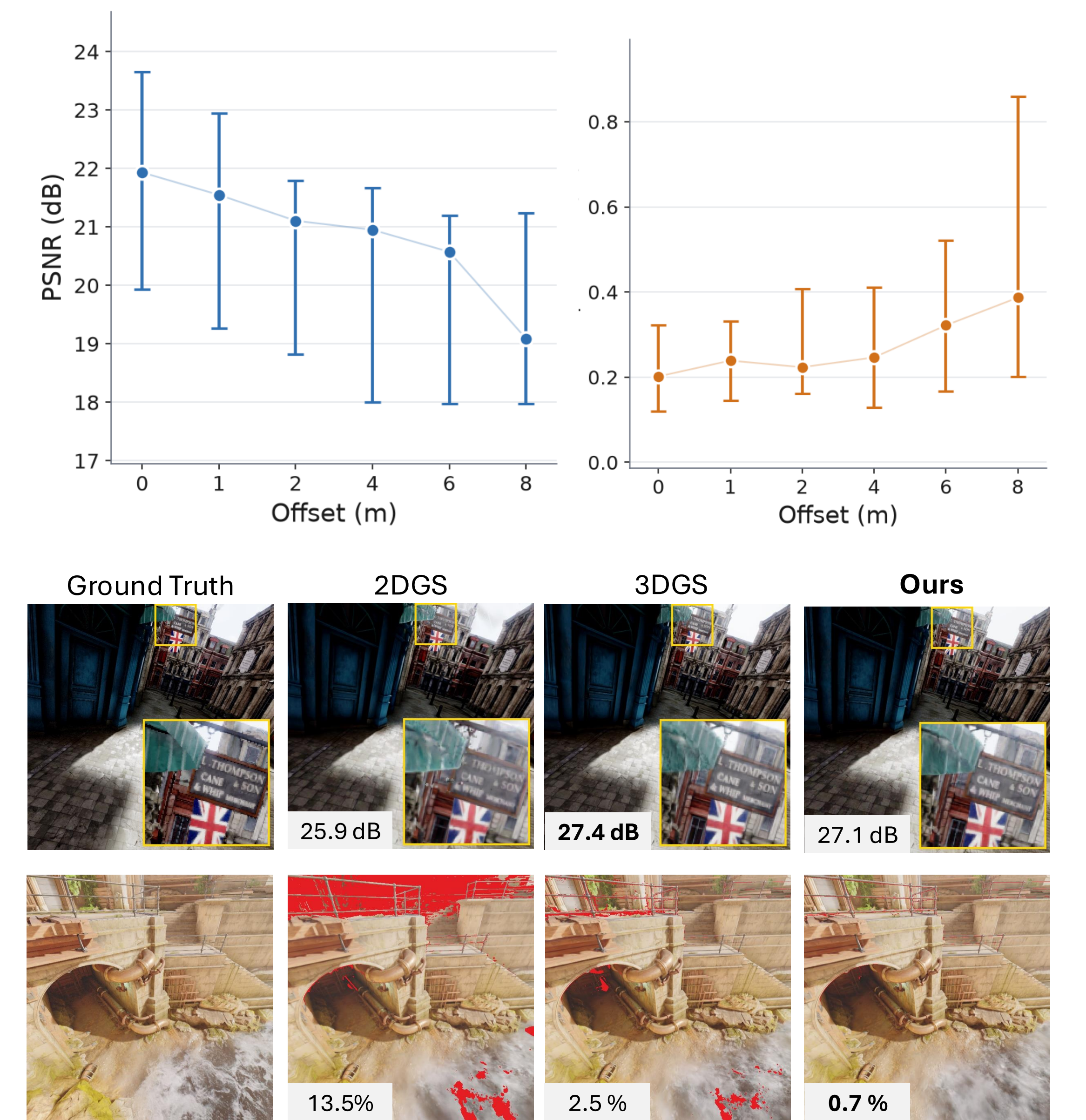
Imitation Learning inside 3DGS environment



Results



Qualitative Results



Method	Rendering Quality			Geometric Fidelity	
	PSNR ↑	SSIM ↑	LPIPS ↓	Depth MAE (m) ↓	Floater ↓
Vanilla 3DGS	29.35	0.8706	0.215	0.974	11.5 %
Vanilla 2DGS	29.07	0.8656	0.228	1.466	13.0 %
2DGS + depth	28.93	0.8664	0.224	0.124	0.6%
3DGS + depth (ours)	29.10	0.8729	0.209	0.087	0.4 %

Future Work

- **Evaluate sim-to-real transfer:** Deploy trained RGB policy on a physical quadrotor.
- Extend 3DGS scene reconstruction beyond TartanAir to **broader real-world data**.
- More training regimes beyond imitation learning.

Acknowledgements

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